THE ECB BLOG

# AI adoption and employment prospects

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> *AI is already part of many workers' daily routines. Some fear losing their jobs, but most don't. The ECB Blog looks at how workers are using AI tools, how they feel about it and what that means for work in the future.*

**This post is part of a miniseries related to the ECB conference "The Transformative Power of AI", on 1-2 April 2025, bringing together researchers, practitioners and policymakers. Learn more here.**

The dawn of AI was accompanied by promises of new jobs being created and workers, including those at the bottom of the earnings distribution, benefitting from better work.[1] But there were also fears about nearly every job being automated, making many workers completely redundant. In this post we use the ECB's Consumer Expectations Survey (CES) to look at how workers *actually* perceive AI and what effect they expect it will have on them. We also investigate how personal or job characteristics affect those views.

## How many workers use AI in their jobs?

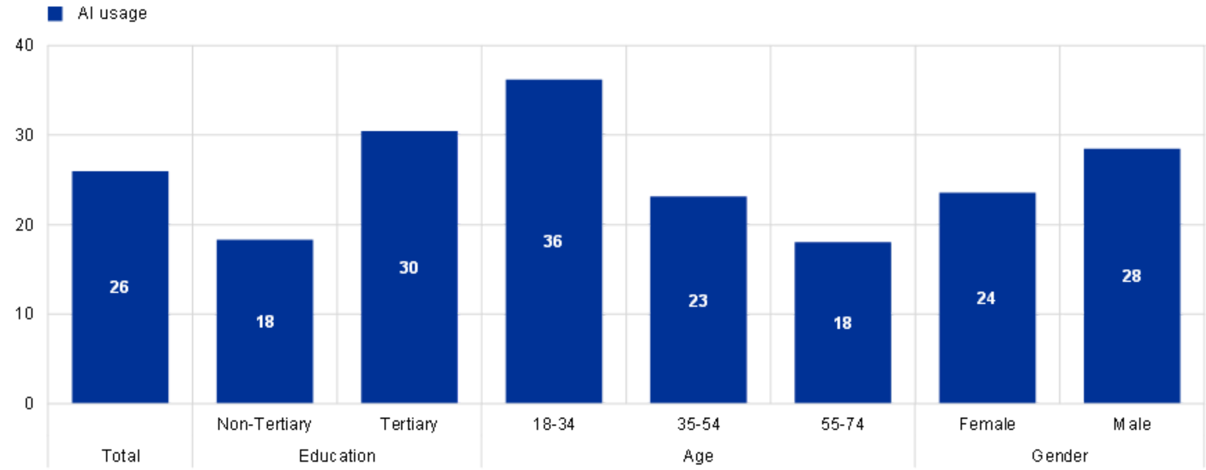
The May 2024 CES asked workers in 11 euro area countries whether they used AI at work.[2] About a quarter responded that they did in fact use AI tools.[3] As shown in Chart 1, there were some interesting patterns in the responses. First, there is a notable age bias. Workers aged 18-34 are twice as likely to use AI in their work (36%) as those aged 55-74 (18%). Second, workers with university educations are also more likely to use AI (30%) than those without (18%). Third, there is a less pronounced but noticeable gender difference, with men more likely to use AI (28%) than women (24%).[4]

AI usage also varies across activity sectors and workers' occupations, as we will discuss below. And that has implications for how workers' perceive the impact of AI on their jobs.

**Chart 1**

AI usage by demographic group

Percentage of workers




Sources: ECB Consumer Expectations Survey. Notes: Share of employed workers reporting the use of AI in their work. The question reads: "Do you personally use artificial intelligence (including a large language model, such as ChatGPT or Gemini) in your work?".

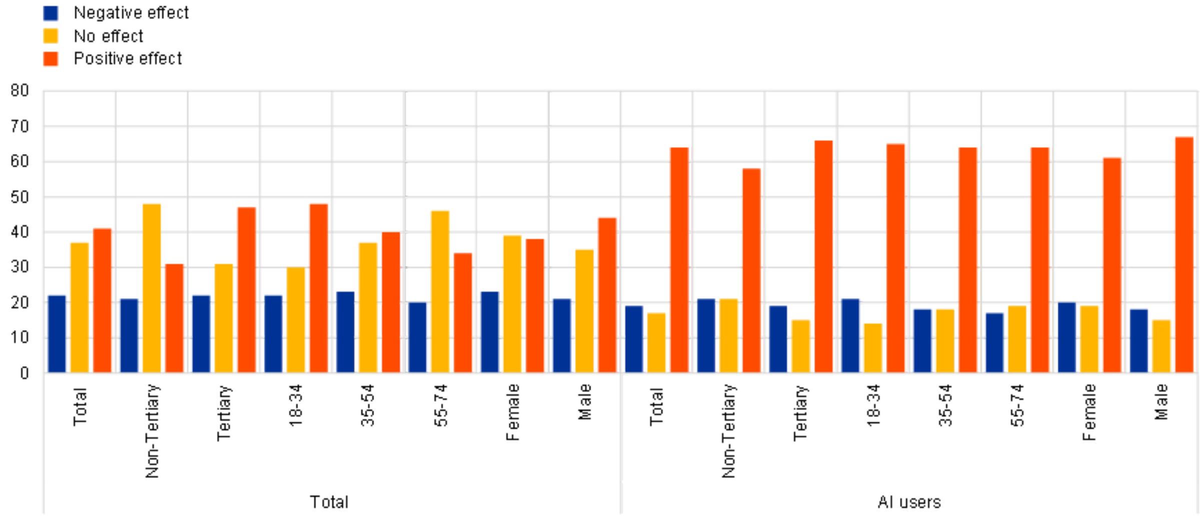
## What do workers think about AI?

The CES also asked workers about their expectations regarding the impact of technological advancements like AI on their current jobs and their employment prospects over the next five years. Most workers expressed neutral or favourable views. Approximately 41% of all workers believe that new technologies will have a positive effect on their productivity or job opportunities (Chart 2). Another 37% do not anticipate any effect. This means that only about 20% of workers have negative expectations, possibly fearing job losses or worsening employment prospects. Interestingly, there are again visible differences across demographic groups. We find significantly more positive views of AI among workers with university education and younger workers in particular. These differences may be related to the actual usage of AI. As we described above, it is exactly these younger and university educated workers who use AI more frequently in their jobs already. Overall, those who use AI tools tend to have more favourable views than those who don't. Nearly two-thirds of them foresee a positive effect on their future employment prospects. This may be because users with a positive view of AI tools are more likely to use them, or because using these tools has helped them discover the potential benefits.

## Chart 2

AI usage and perceptions of impact on jobs

Percentage of workers




Sources: ECB Consumer Expectations Survey.

Notes: The question reads "Do you think that technological advancements (e.g. the increasing use of artificial intelligence, large language models, automation) will affect your current job or employment prospects in the next five years?" with three possible answers: a) Yes, they will affect me positively (e.g. by increasing my productivity and/or improving my employment prospects); b) Yes, they will affect me negatively (e.g. by making my job less necessary and/ or worsening my employment prospects); and c) No, they will not affect me at all.

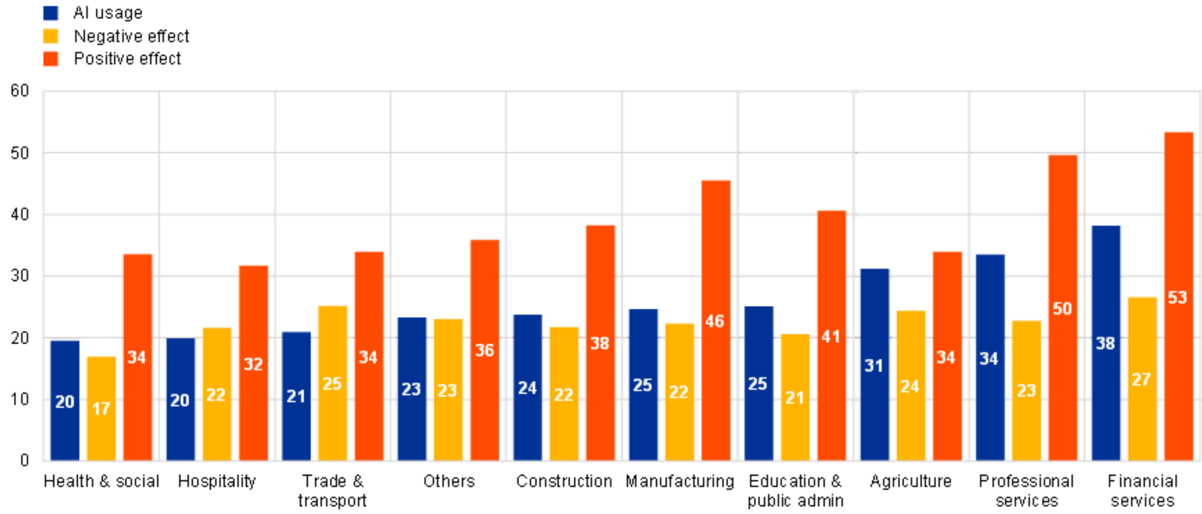
AI use varies from one sector to another (Chart 3). The share of workers using AI is highest in the financial services sector, followed by professional services. At the other end of the scale are workers in the health & social, health care and trade & transport sectors.

Much like demographic groups, those sectors with the highest use of AI tools also report the most positive attitudes towards them (Chart 3). Conversely, the sectors with the lowest shares of AI use report some of the least positive attitudes. These disparities may reflect the fact that workers in some sectors see AI as more of a complement to their work, while workers in others sectors instead fear that AI could replace them. Workers in sectors characterised by manual tasks appear to have less favourable attitudes towards AI, while those involving analytical tasks report more favourable views. Meanwhile, negative perceptions remain strikingly consistent across sectors. Analysis shows that this negative sentiment is strongly linked to workers' perceived job loss risks. This confirms that the risk of total or partial replacement by technology could be a main driver behind negative attitudes towards AI and similar technologies.

**Chart 3**

AI usage and perceptions of impact across activity sectors

Percentage of workers




Sources: ECB Consumer Expectations Survey.

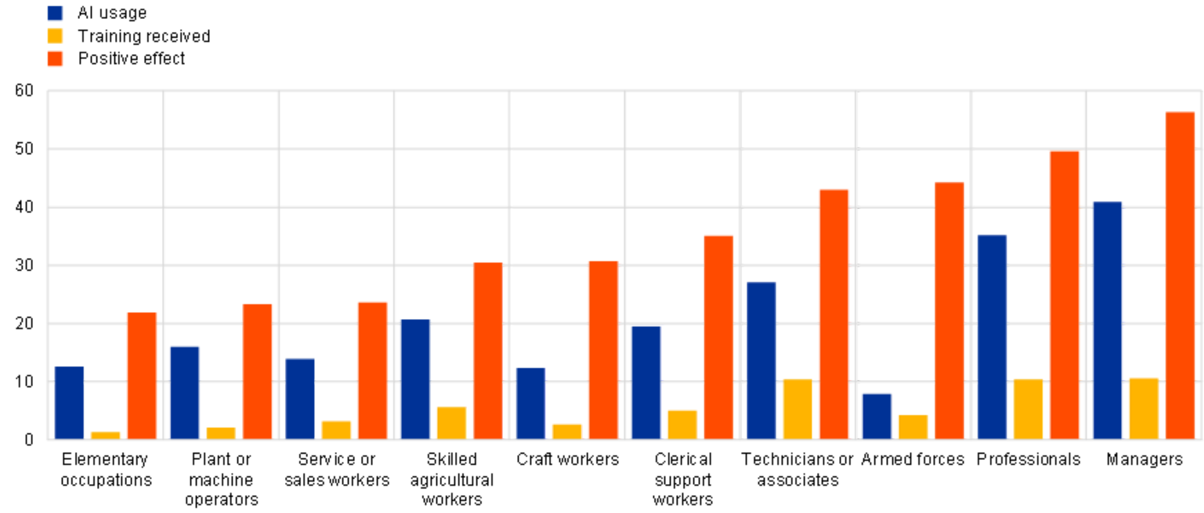
Notes: The shown perception percentages exclude workers who expect no effect of technological advancements (such as AI or automation) on their current or future employment prospects. Perceptions are reported for all workers independently of their AI usage.

Managers, professionals and technicians are employee groups that are more likely to integrate AI into their daily tasks, have more positive attitudes towards it, and believe AI will positively impact their jobs (Chart 4). Notably, more than half of all managers view AI favourably in their occupations. These differences in attitudes towards AI of people working in different occupations could be related to training opportunities. There is a positive link between the amount of training received and the level of AI adoption. This suggests that workers' attitudes towards AI are greatly influenced by their proficiency with these technologies and their ability to harness them to boost productivity and enhance their job market prospects.

**Chart 4**

AI usage and perceptions of impact across occupations

Percentage of workers




Sources: ECB Consumer Expectations Survey. Notes: The shown perception percentages exclude workers who expect either no effect or negative effect of technological advancements (such as AI or automation) on their current or future employment prospects. Perceptions are reported for all workers independently of their AI usage.

## What does this mean for the future?

New technologies often spark fears of job losses. And those fears are not completely unfounded as whole professions could become obsolete. But at the same time those technologies have in the past generally created even more new tasks and opportunities. It remains to be seen whether AI will have the same net positive effect. At the moment, workers in the euro area have diverging views on the topic, with those already using AI tools generally having significantly more positive views, suggesting that familiarity reduces fear. Data show that familiarising workers with new AI technologies can improve their perceptions and attitude towards them. The CES remains a good barometer of the effects of AI on euro area workers, and we will continue to monitor this as technologies develop.

*The views expressed in each blog entry are those of the author(s) and do not necessarily represent the views of the European Central Bank and the Eurosystem.*

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1.

Evidence from the euro area for the period 2011-2019, marked by great progress in machine learning and deep learning technologies, shows that occupations more exposed to AI saw an increase in their employment shares, suggesting potential benefits: Albanesi, S., Dias da Silva, A., Jimeno, J. F., Lamo, A., and Wabitsch, A. (2025). “New technologies and jobs in Europe.” *Economic Policy*, 40, 121: 71–139. Additionally, for the potential equalising effect of AI see Brynjolfsson, E., Li, D., and Raymond, L. R.

(2023) "Generative AI at Work." *Quarterly Journal of Economics*, qjae044, https://doi.org/10.1093/qje/qjae044.

2.

These are: Belgium, Germany, Ireland, Greece, Spain, France, Italy, Netherlands, Austria, Portugal and Finland.

3.

As the CES is an online survey the share of workers using AI may be higher here than in other surveys that are not administered online.

4.

The patterns across demographic groups are consistent with evidence in other surveys. For example, using the Federal Reserve Bank of New York Survey of Consumer Expectation (SCE), the BIS reports that AI usage is lower for women, persons without college degree and persons older than 60 in the United States as well. See Aldasoro, I., Armantier, O., Doerr, S., Gambacorta, L., and Oliviero, T., “Survey evidence on gen AI and households: job prospects amid trust concerns” *BIS Bulletin*, No 86, 23 April 2024.